

Custody

Proving digital evidence has not been changed

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The problem

A flash drive. An email. A shared laptop.

A hearing, eleven months later.

**Nobody can prove the file is
the one that was collected.**

Not because anyone changed it. Because nobody can show they did not.

Where this happens

Three settings

Cybercrime cases

phone extractions, server logs

Fraud investigations

transaction records, audit logs

Disciplinary panels

message exports, access logs

In each one the handling record is on paper, if it exists at all,
and the evidence is a file that anyone could have opened.

The idea

Same file, same code. Every time.

statement.pdf

A3F1 9C22 7E04 6B8D 1C55 E210 44AF 9D31

statement.pdf (one character changed)

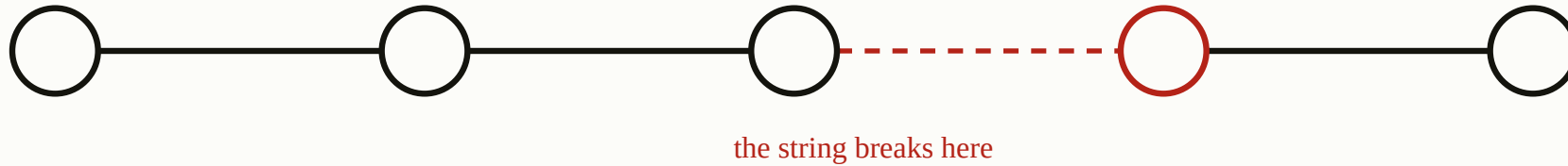
8B02 41DE C5A7 F014 92E6 3B7C D885 07A2

One character changed in a 40 page statement.

Nothing else about the file is different.

The idea, part two

Every entry is knotted to the one before it, like beads on a string.



You cannot quietly remove a bead from the middle.

How it fits together

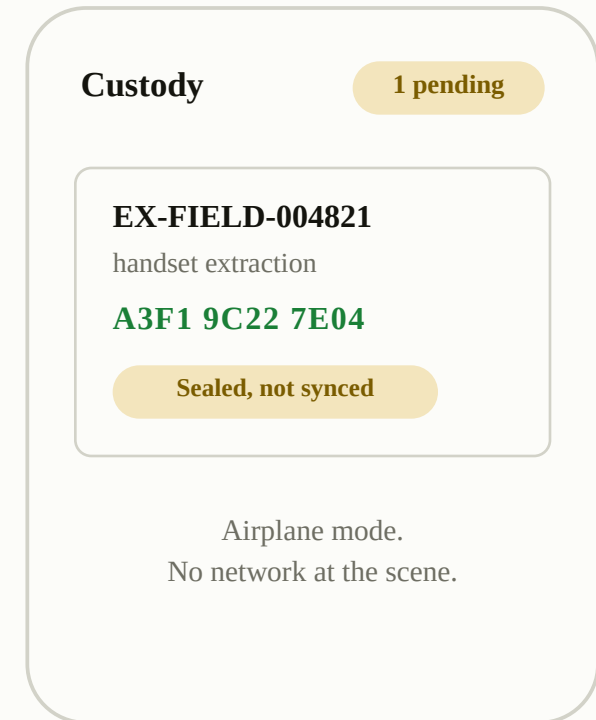
Architecture



One laptop, or hosted. Nothing paid for. Works with the network down.

The fingerprint is taken at the scene, on the officer's own device, before the file goes anywhere.

Nothing between the scene and the server can alter it undetected.



One extraction can exceed 100GB

**We say which part changed,
not just that something did.**



One byte altered in thirty million. Part 6 of 8 differs, so the change lies between byte 20,971,520 and byte 25,165,823.

Demo

One page. No jargon. No hashes.

Is it unchanged?

Yes.

How do we know?

A fingerprint was taken at collection, and checked again.

Who has handled it?

Four people, named, with dates.

Could the list be edited?

No, and here is why.

A process that is technically perfect but cannot be explained in a hearing has not solved the problem.

the brief

We tested it on people

**We gave the report to someone
outside computer science.**

What confused them [to complete]

What we changed [to complete]

What we did not solve

Honest limits

A corrupt collector

protected from collection onward, not before it

The device clock

both clocks shown, the phone's not vouched for

Scale

tested at 30MB, extrapolated, not a real extraction

Per officer signing

tied to a login, not to a person's key

Storage

change is proven, deletion is not prevented

Bulk compromise

single edits are caught, a full rewrite is not

If asked

Why not a blockchain?

A blockchain solves distrust between parties with no shared authority. A court has one.

A hash chain and append only storage give the same tamper evidence, and can be explained to a panel in one sentence.